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LABORATÓRIO DE SÍNTESE EM
BIODIVERSIDADE
Universidade Federal de Mato Grosso do Sul



Alexander von
HUMBOLDT
STIFTUNG

Evolutionary Macroecology of the Atlantic Forest of South America

Prof. Dr. Diogo B Provete

Assistant Professor – Fed Univ Mato Grosso do Sul (Brazil)

Sabbatical at iDiv (Leipzig, Deutschland)

Experienced Researcher avHumboldt Stiftung



iDiv

German Centre for Integrative
Biodiversity Research (iDiv)
Halle-Jena-Leipzig

- My wanderings in between community ecology and evolutionary biology
- Metacommunity ecology
- Link to biogeography
- Formation of the regional pool
 - Evolution of ecologically-relevant traits at the community scale

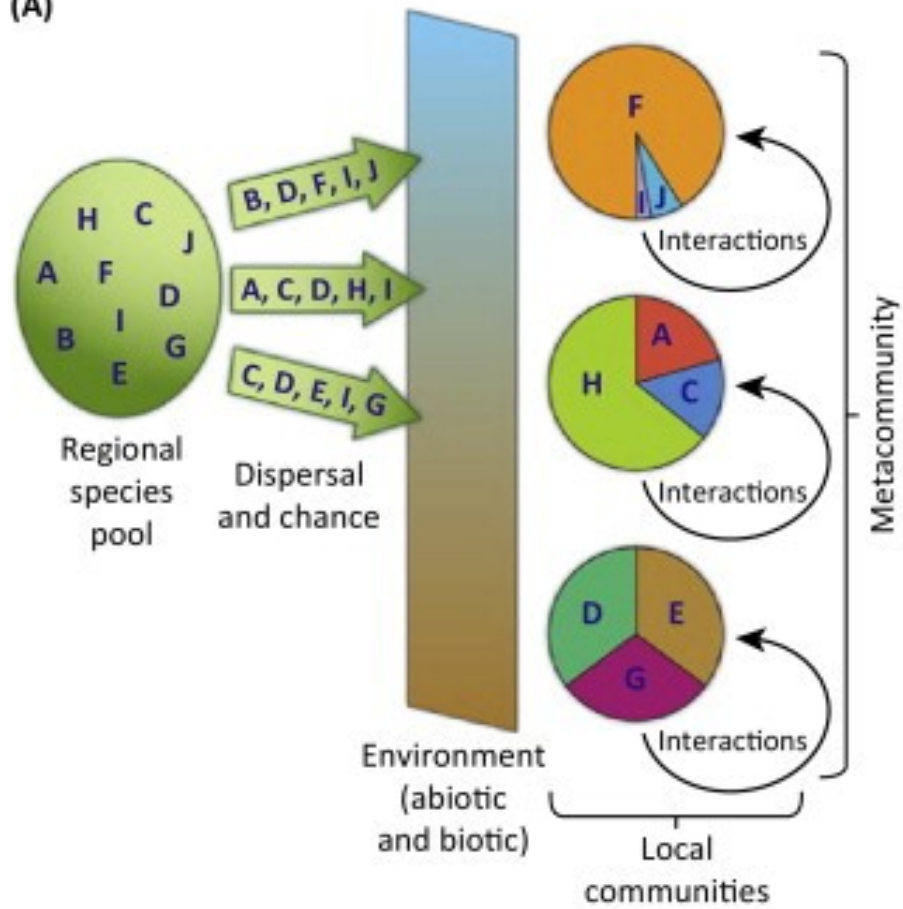


Serra da Bocaina National
Park, SE Brazil, 2008

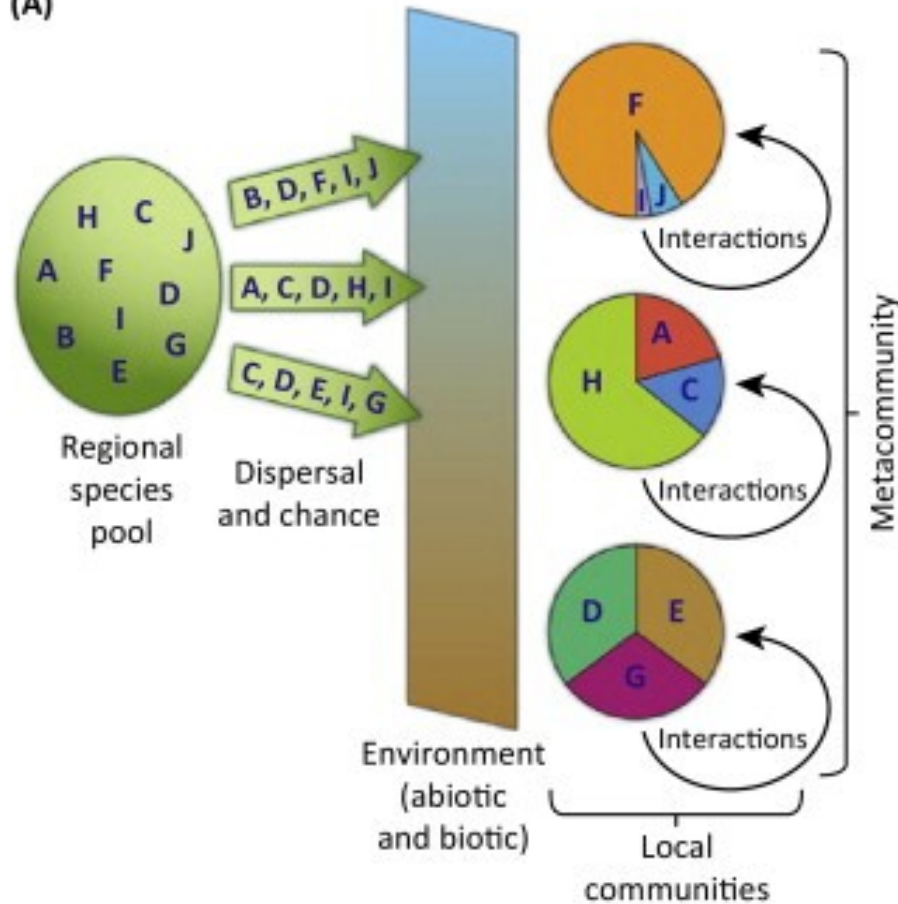
Tadpole
s!



(A)

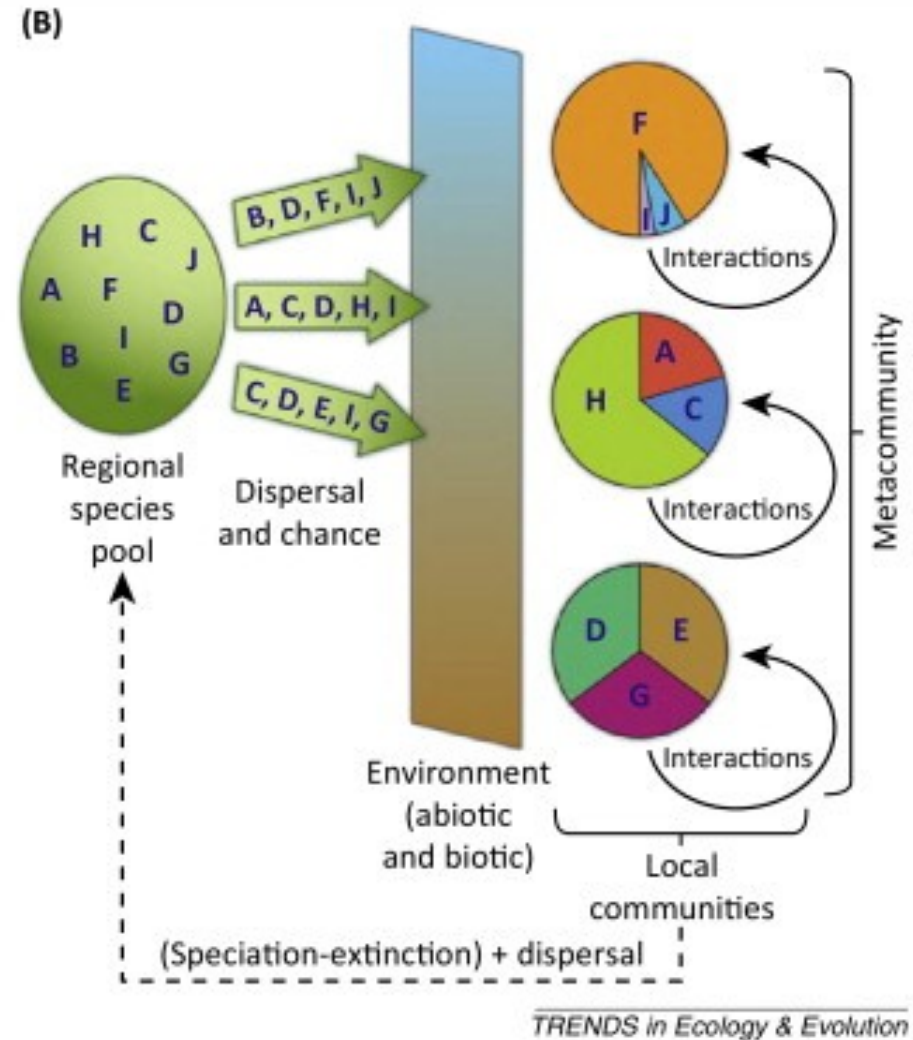


(A)



But something
was missing...

Understanding how the regional species pool came to be is as important as understanding local community assembly through hierarchical filters



Review

**Biogeography and ecology: towards
the integration of two disciplines**

Robert E. Ricklefs^{1,*} and David G. Jenkins²

VOL. 172, NO. 6 THE AMERICAN NATURALIST DECEMBER 2008

Disintegration of the Ecological Community

American Society of Naturalists Sewall Wright Award Winner Address^{*}

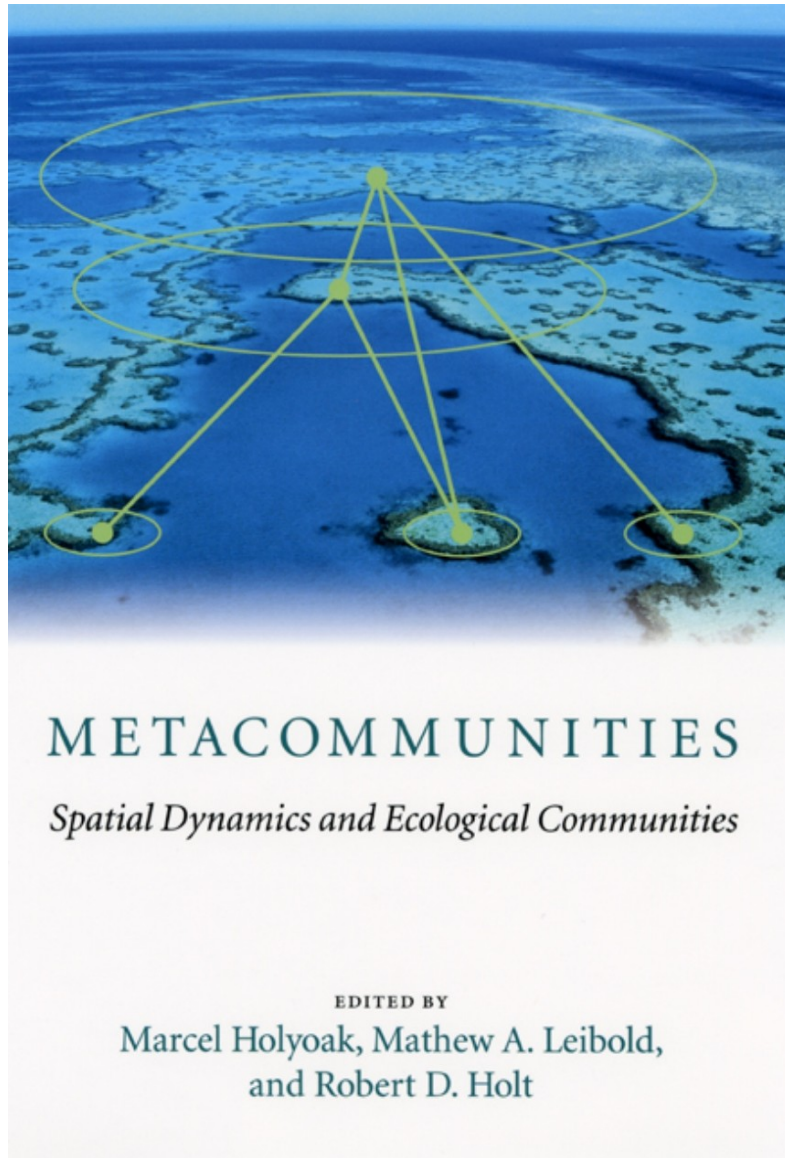
Robert E. Ricklefs[†]

Articles

**Community Diversity: Relative Roles of
Local and Regional Processes**

ROBERT E. RICKLEFS

1987



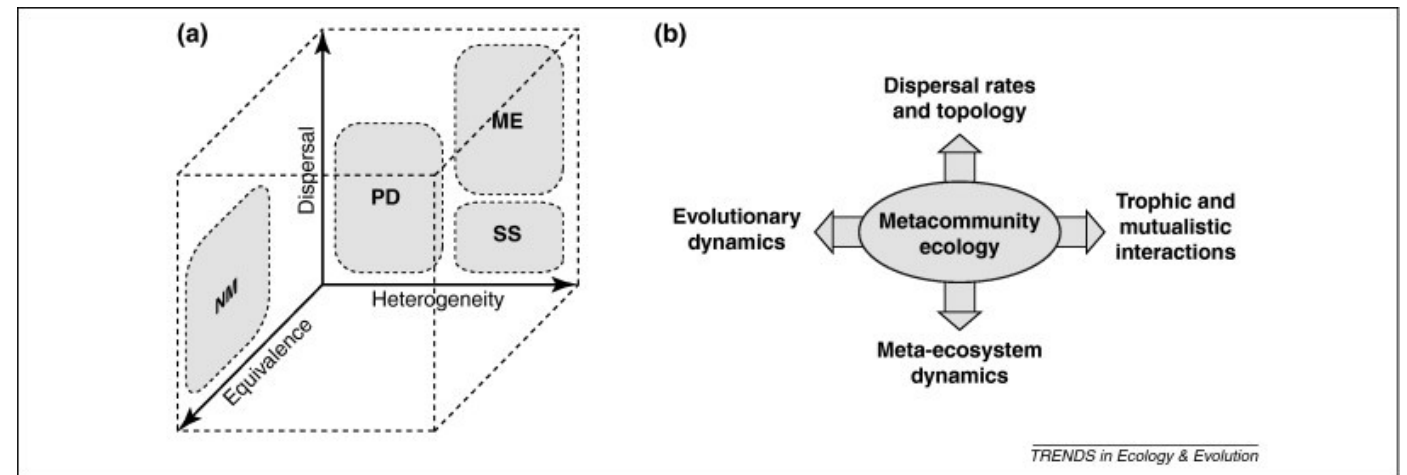
REVIEW

The metacommunity concept: a framework for multi-scale community ecology

Abstract

The metacommunity concept is an important way to think about linkages between different spatial scales in ecology. Here we review current understanding about this concept. We first investigate issues related to its definition as a set of local communities that are linked by dispersal of multiple potentially interacting species. We then identify four paradigms for metacommunities: the patch-dynamic view, the species-sorting view, the mass effects view and the neutral view. that each emphasizes different processes of

M. A. Leibold,^{1*} M. Holyoak,²
 N. Mouquet,^{3,4} P. Amarasekare,⁵
 J. M. Chase,⁶ M. F. Hoopes,⁷
 R. D. Holt,⁸ J. B. Shurin,⁹ R. Law,¹⁰
 D. Tilman,¹¹ M. Loreau¹² and
 A. Gonzalez¹³



Ponds sampled (13)

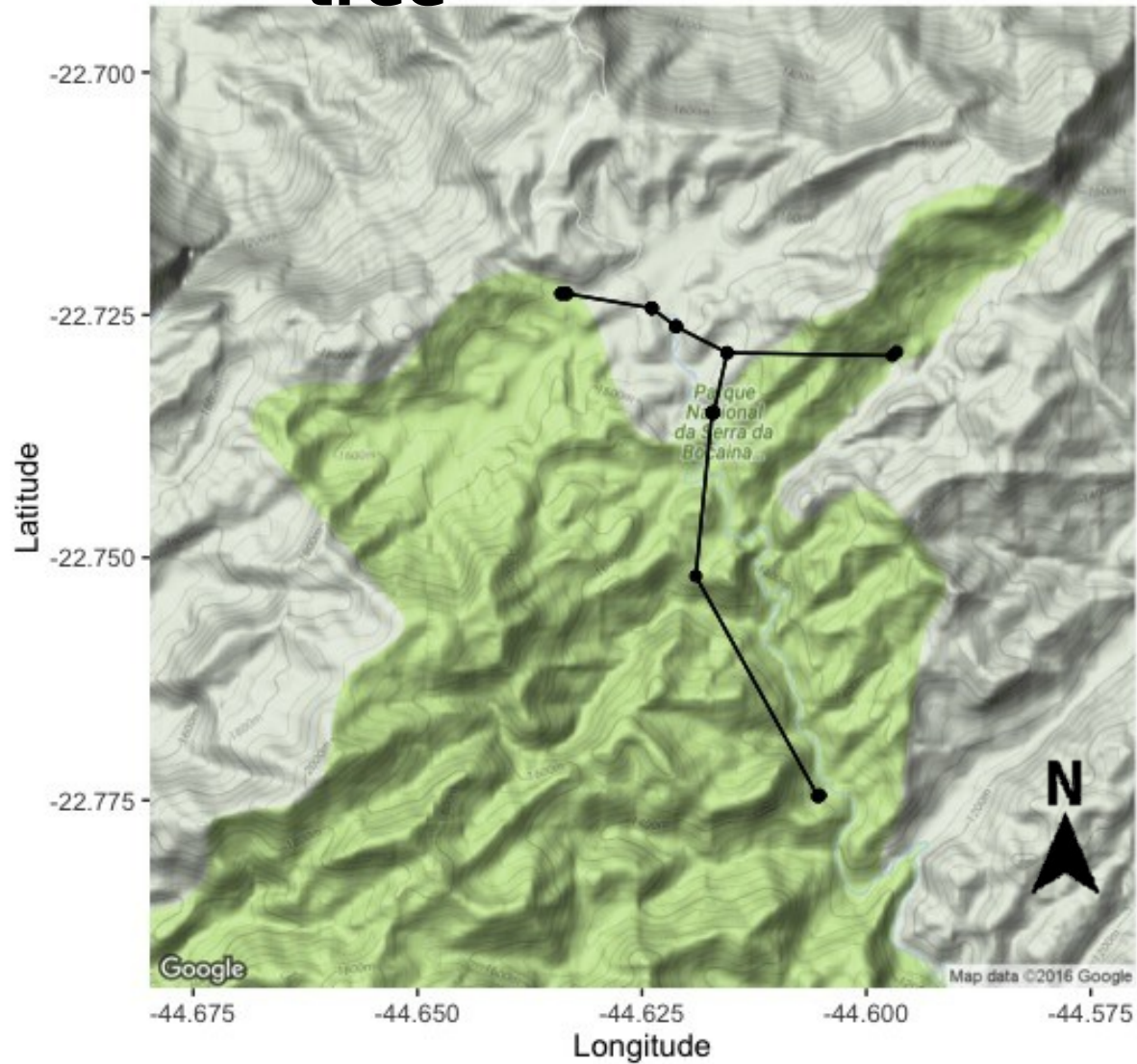


Environmental data

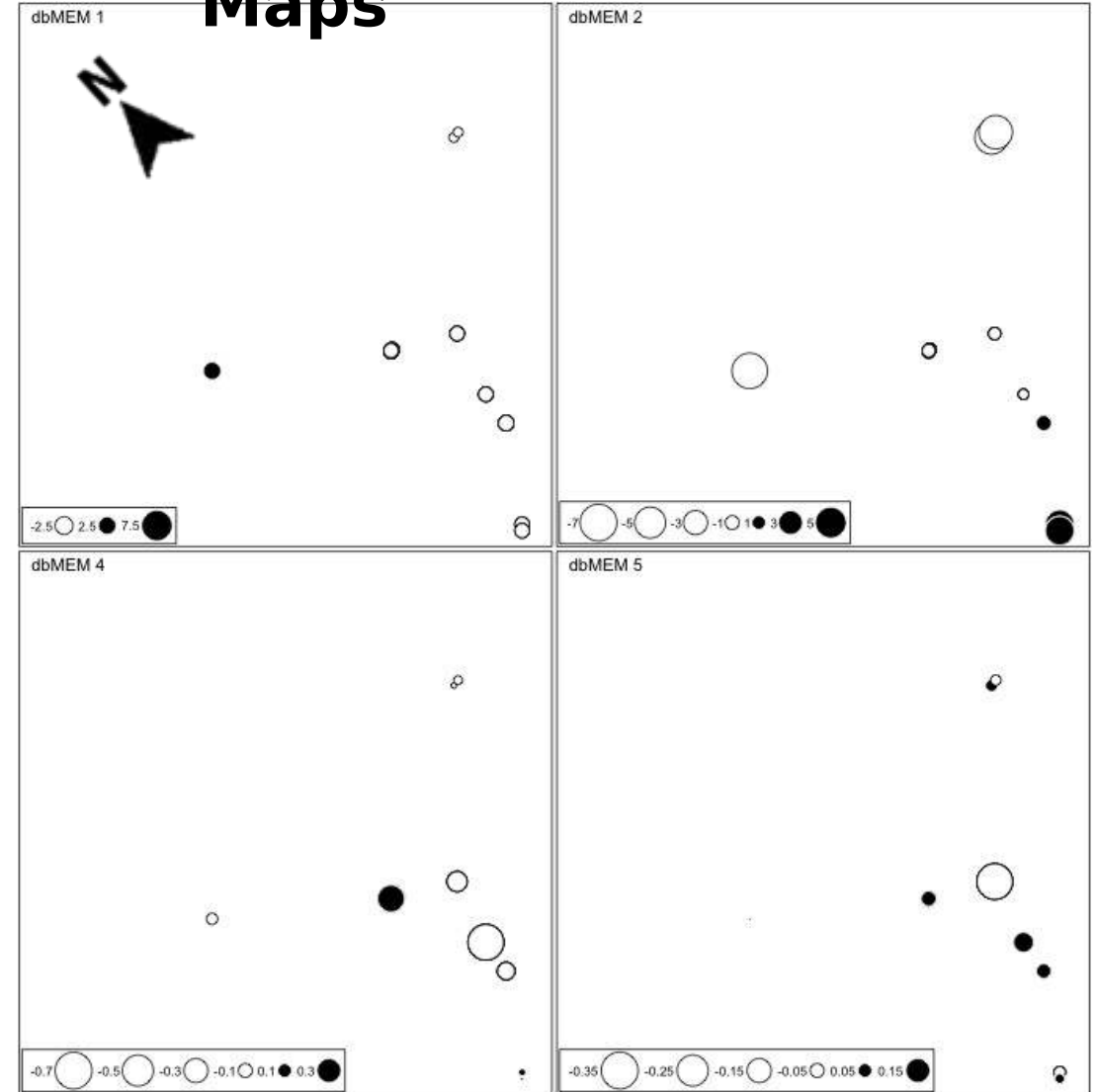
- Water chemistry (pond scale)
 - pH, DO, Turbidity, Temperature, Conductivity (Hanna)
- Pond morphology and landscape variables
 - hydroperiod, area, depth, % floating vegetation, % types of vegetation on the margin (3 states - fuzzy)
 - % pond canopy cover, distance to nearest forest fragment
- Microclimate
 - Air temperature and humidity

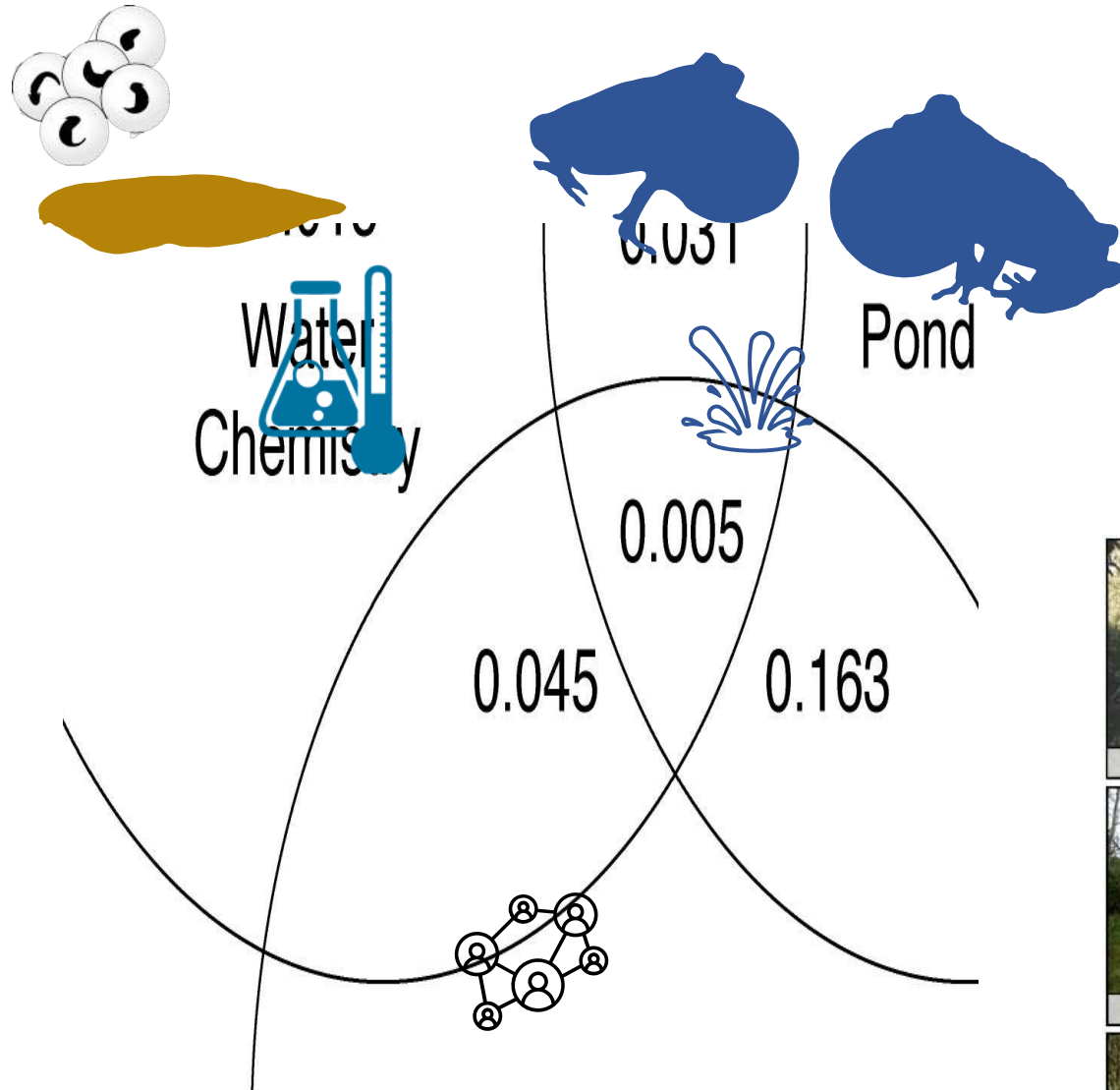


Minimum spanning tree



Moran Eigenvector Maps





Broad-scale spatial patterns of canopy cover and pond morphology affect the structure of a Neotropical amphibian metacommunity

Diogo B. Provete · Thiago Gonçalves-Souza ·
Michel V. Garey · Itamar A. Martins ·
Denise de C. Rossa-Feres

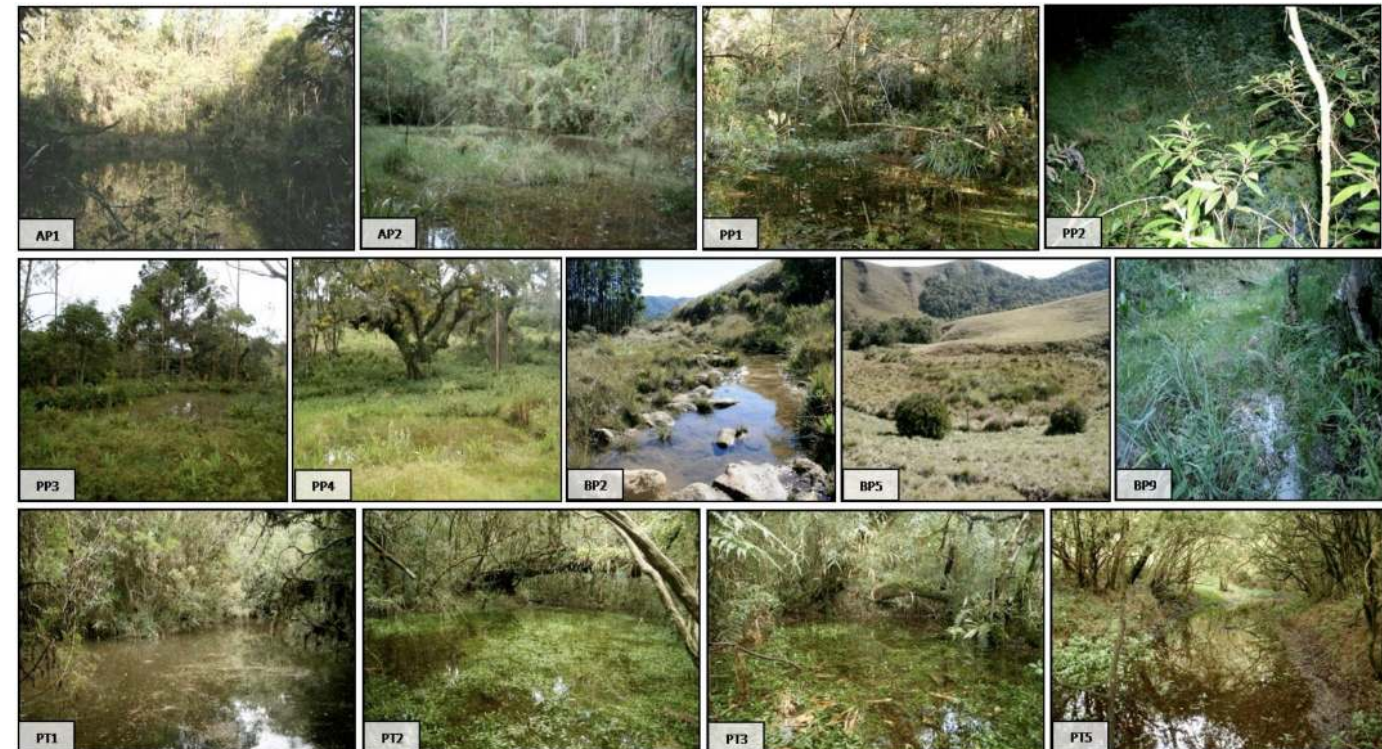
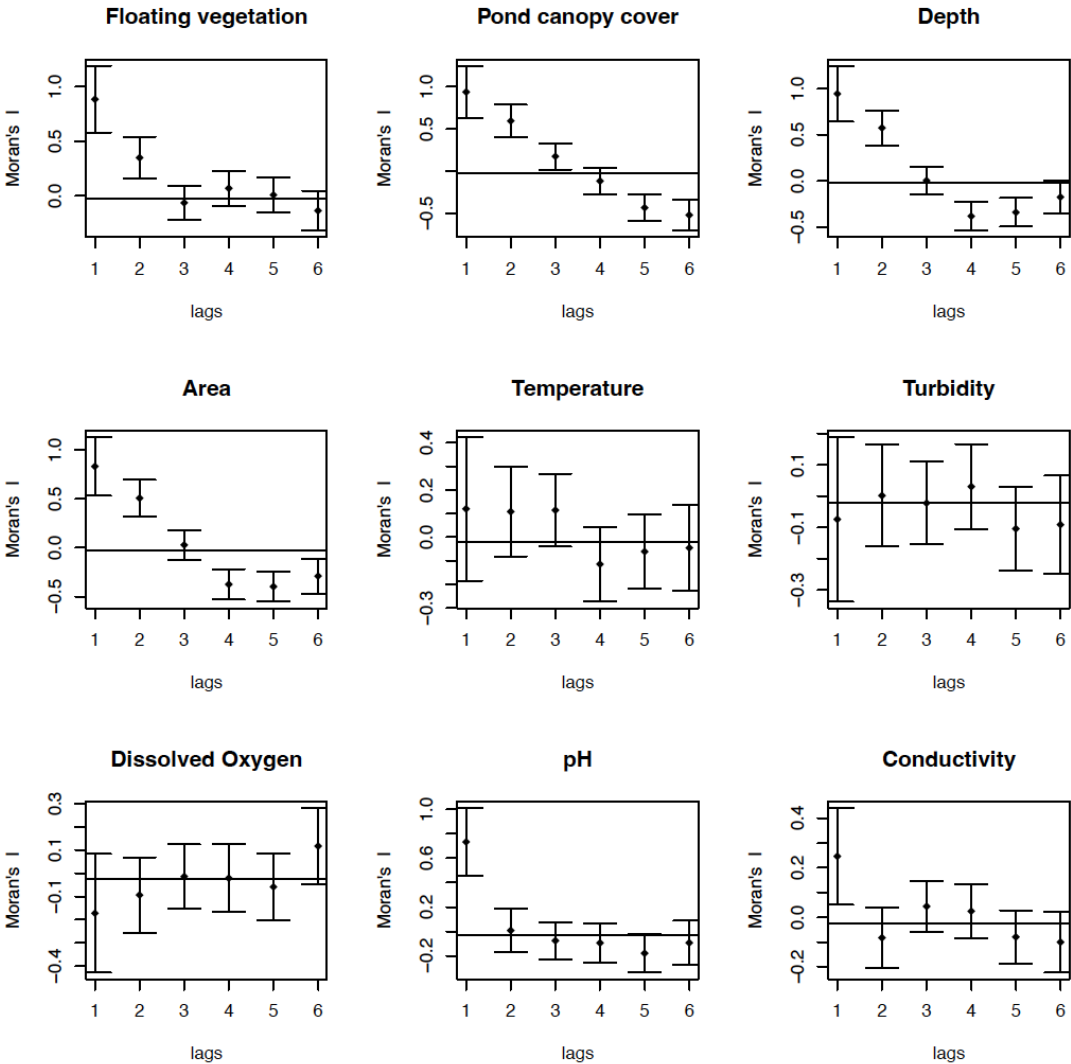


Table 2 Standardized regression coefficients for environmental variables used to predict each spatial variable in the linear model

	dbMEM 1	dbMEM 2	dbMEM 4	dbMEM 5	dbMEM 7
Temperature	−0.170	0.007	−0.164	0.104	−0.352
pH	−0.118	0.032	0.035	0.218	0.131
Dissolved oxygen	− 0.195*	0.033	−0.136	0.177	−0.248
Turbidity	−0.100	0.054	−0.061	0.122	−0.118
Conductivity	0.138	− 0.317**	− 0.191*	0.010	0.016
Floating vegetation	0.254*	− 0.577**	0.259	−0.274	−0.020
Canopy cover	0.176	− 0.565**	0.175	0.046	− 0.433*
Depth	− 0.542**	− 0.281**	− 0.570**	− 0.435*	0.436*
Area	1.037**	−0.251	− 0.233*	0.282	−0.180
R^2	0.85**	0.71**	0.79**	0.36	0.48*

Pond morphology
variables are structured in
space at broad scales,
contrary to water
chemistry ones



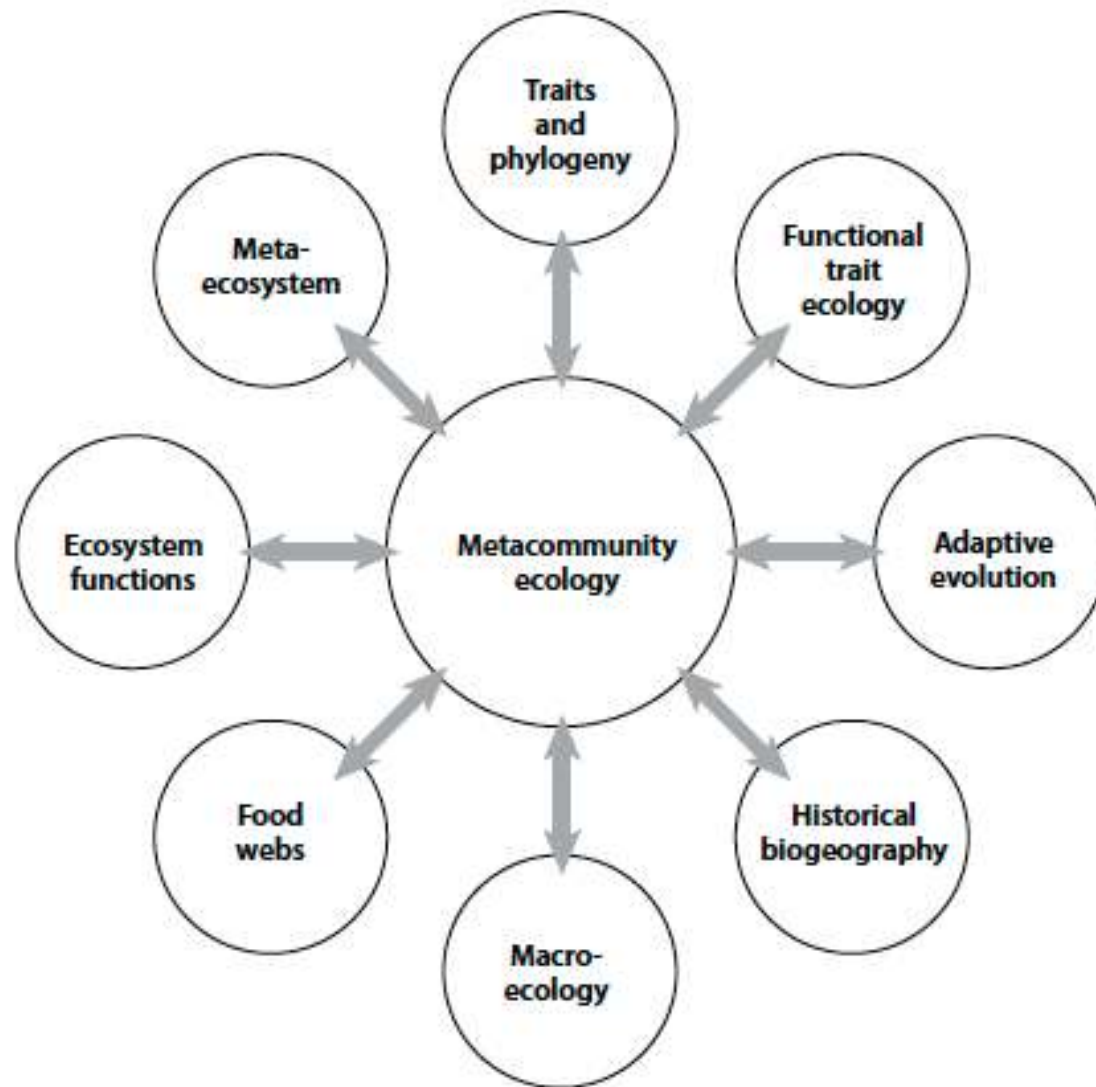


FIGURE 1.6. Metacommunity as a conceptual hub for synthetic ecology. We view metacommunity as both a central concept at the interface of ecological subdisciplines (circles) and as a major axis for novel insights into each subdiscipline (arrows). Shown are the various areas of ecology we address in the book, which correspond to Chapters 6–13, clockwise from the top.

Metacommunity Ecology

Mathew A. Leibold
and
Jonathan M. Chase

MONOGRAPHS IN POPULATION BIOLOGY • 59

Biogeographic Patterns of South American Anurans

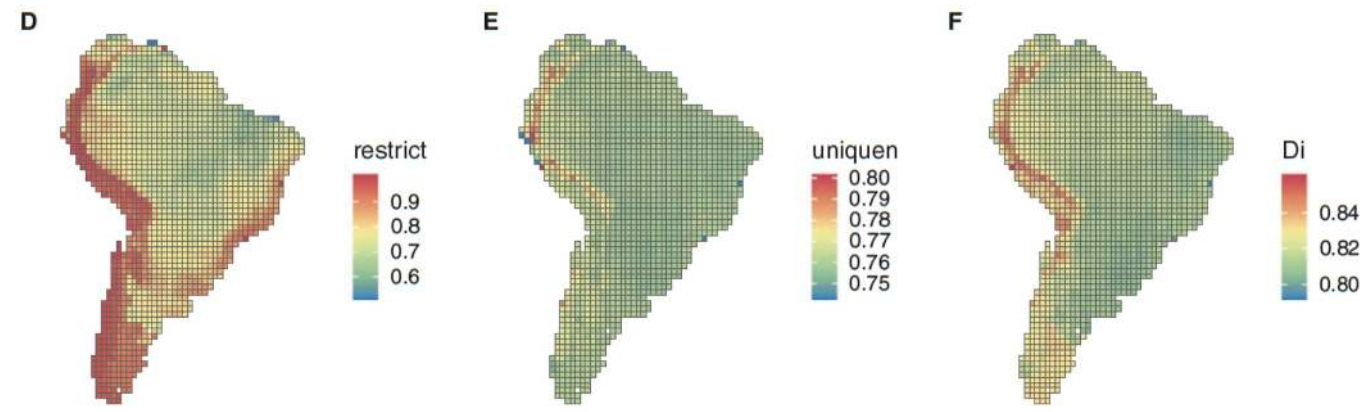
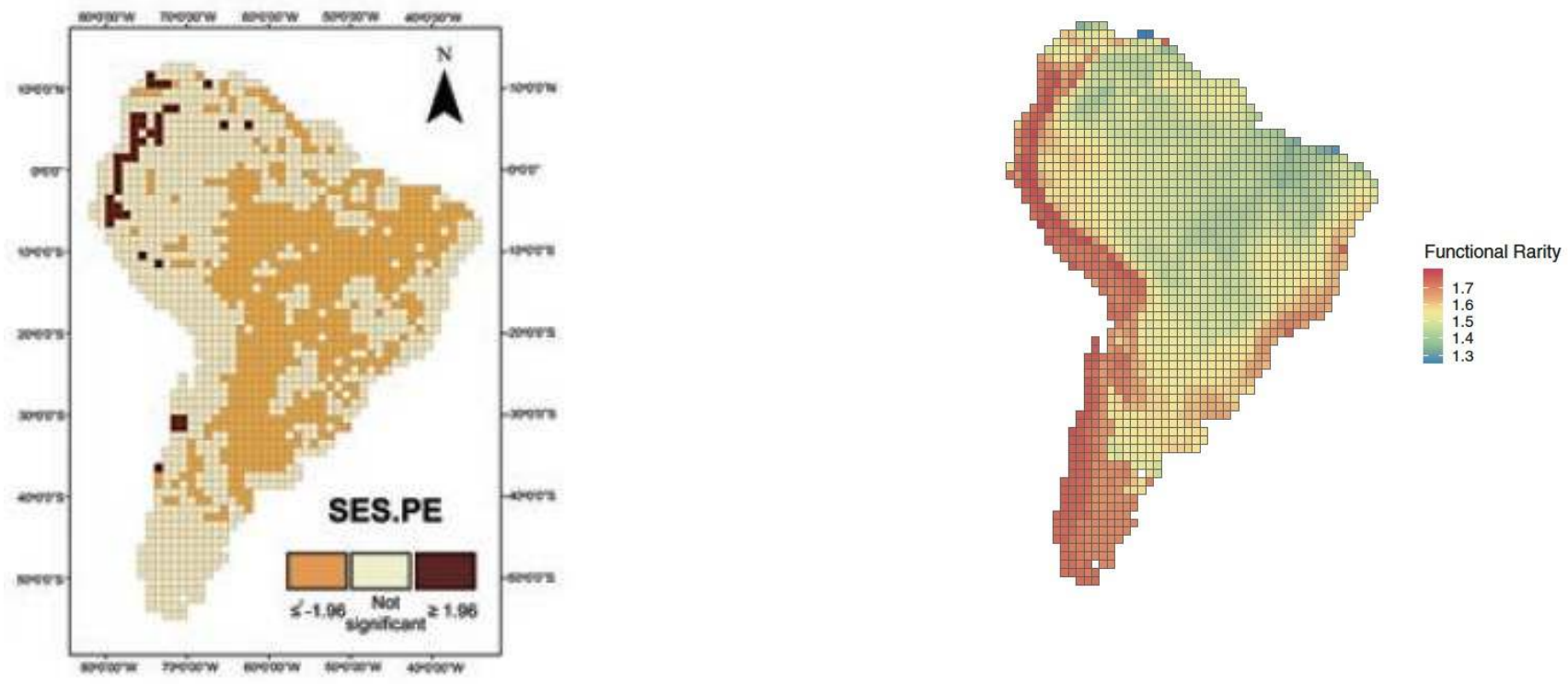


Fig. 5.1 Geographical patterns of functional diversity and functional rarity of anuran amphibians in South America. (a) functional richness, (b) unweighted unctional evenness, (c) functional dispersion, (d) geographical restrictiveness, (e) functional uniqueness, (f) functional distinctiveness



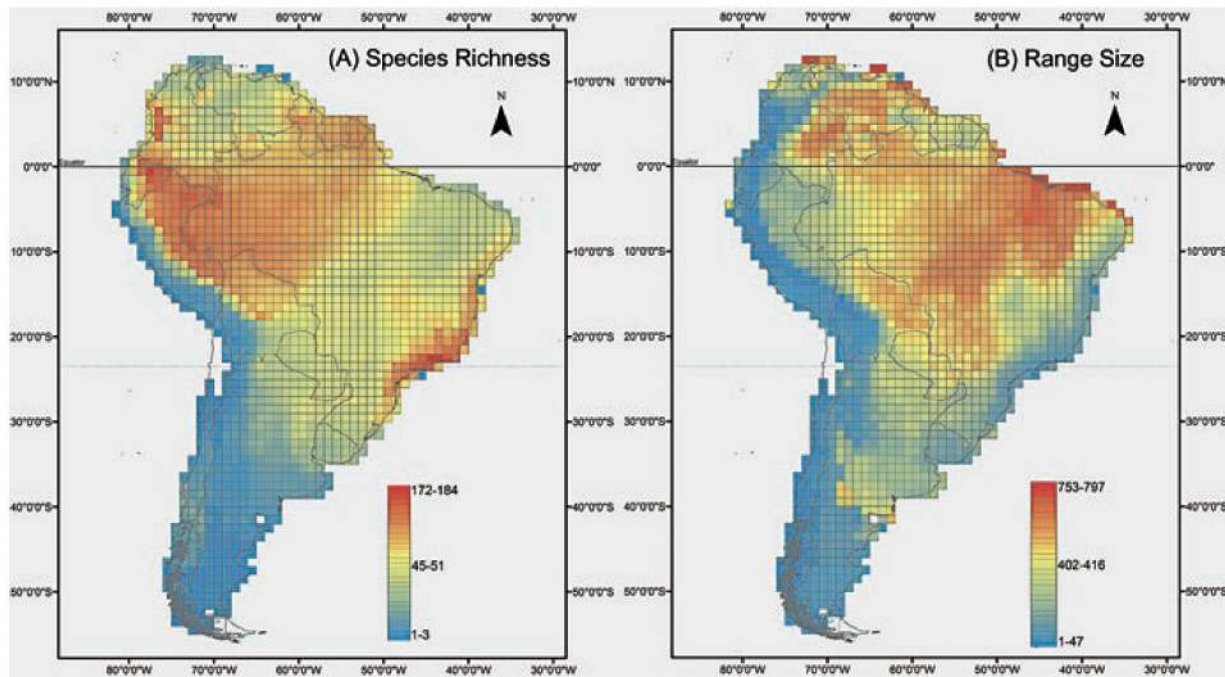


Fig. 3.2 Spatial distribution of anuran (a) species richness and (b) mean range size in 1.0° × 1.0° grid cells across South America

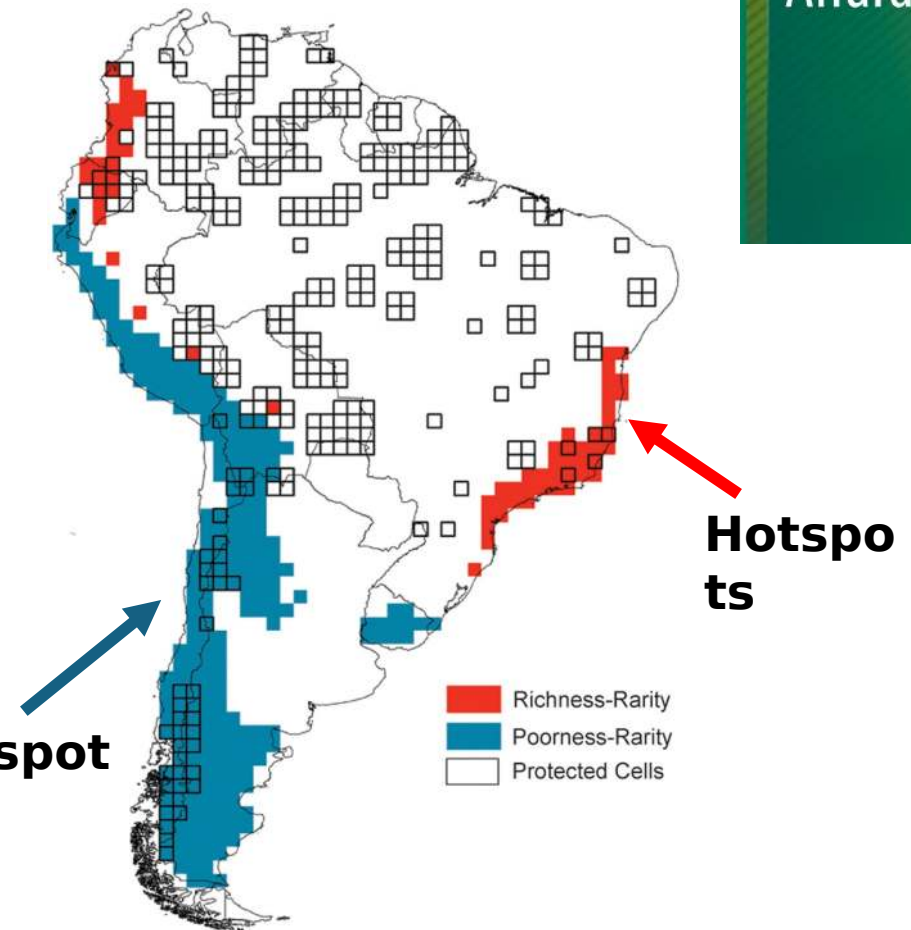
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PLOS ONE

Is Rich and Rare the Common Share? Describing Biodiversity Patterns to Inform Conservation Practices for South American Anurans

Fabricio Villalobos^{1*}, Ricardo Dobrovolski¹, Diogo B. Provete², Sidney F. Gouveia²

Coldspots



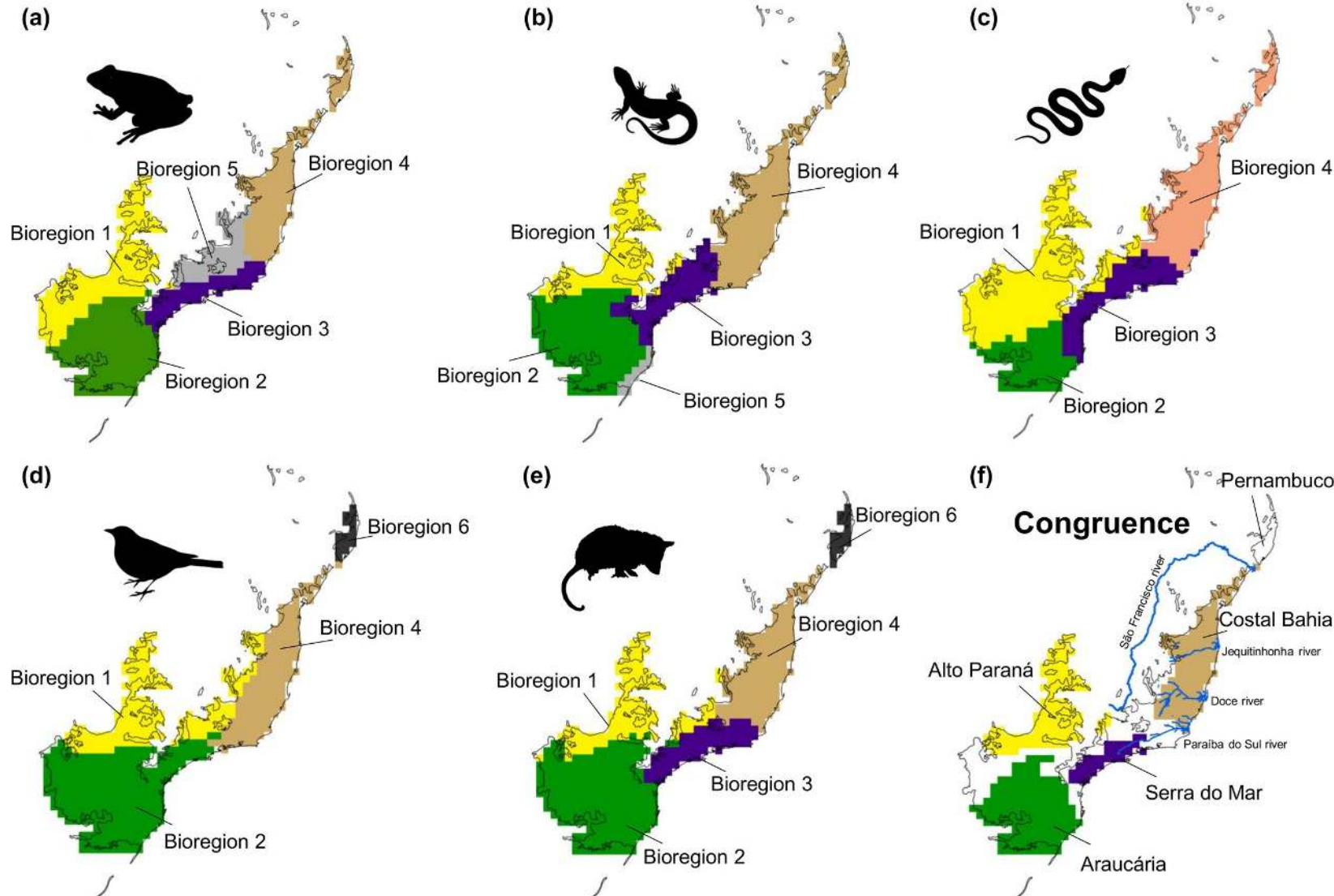
Tiago S. Vasconcelos
Fernando R. da Silva
Tiago G. dos Santos · Vitor H. M. Prado
Diogo B. Provete

Biogeographic Patterns of South American Anurans

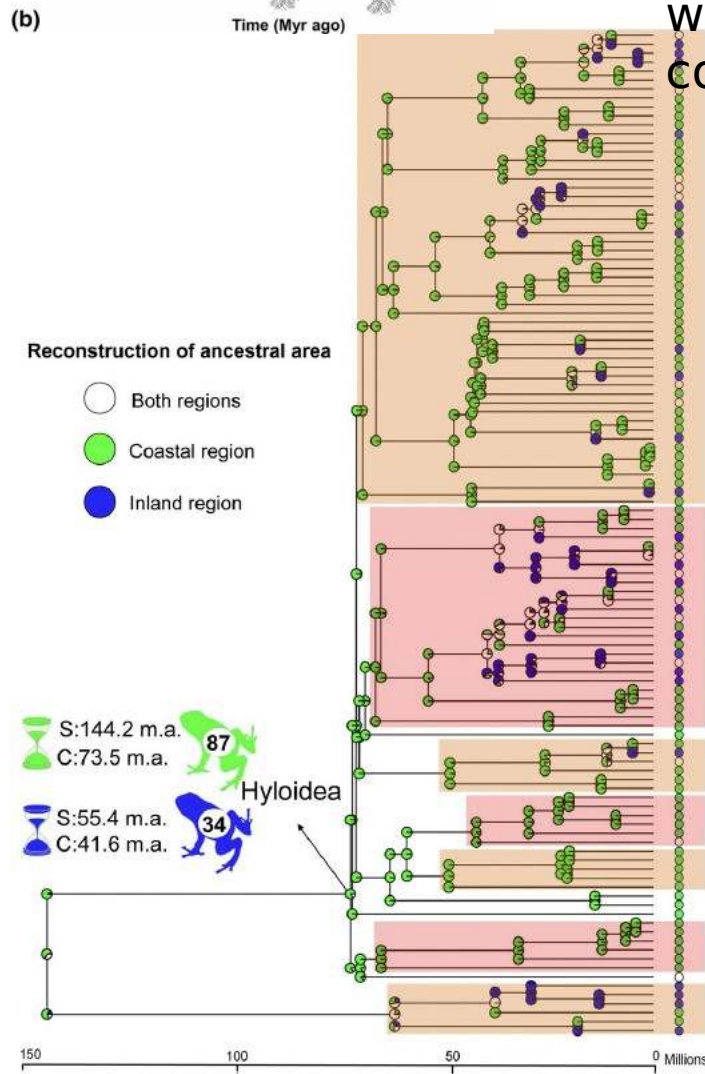
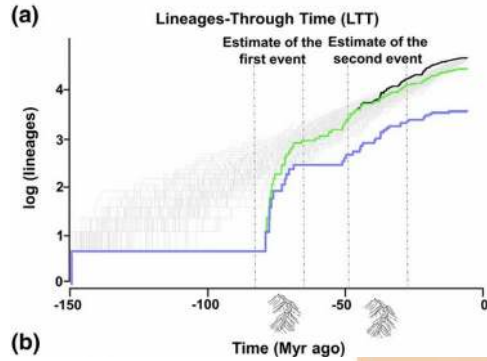
Springer

Zoogeographical regions in the Atlantic Forest: patterns and potential drivers

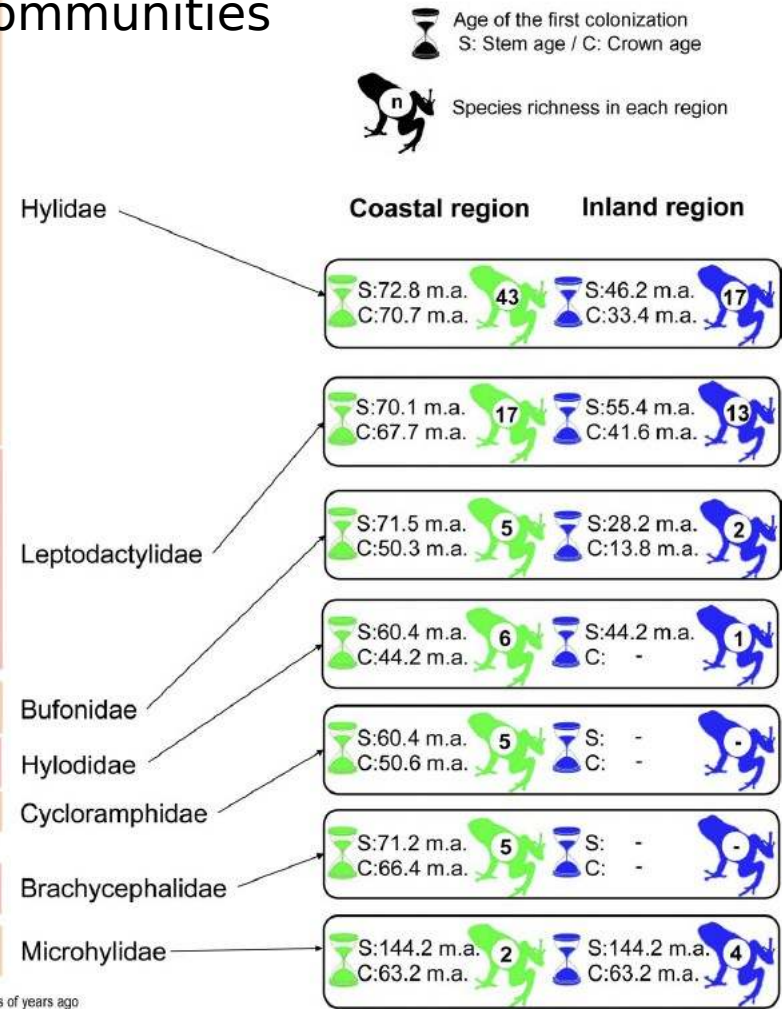
Fernando Rodrigues da Silva¹ | Anna Elizabeth de Oliveira-Silva^{1,2} | Alexandre Antonelli^{3,4,5} | Ana Carolina Carnaval⁶ | Diogo B. Provete^{4,7,8}



- Zooming in the Atlantic Forest we see that regions are more or less congruent between vertebrate groups (when only considering taxonomy)
- Clear distinction between coastal and inland areas in terms of species composition



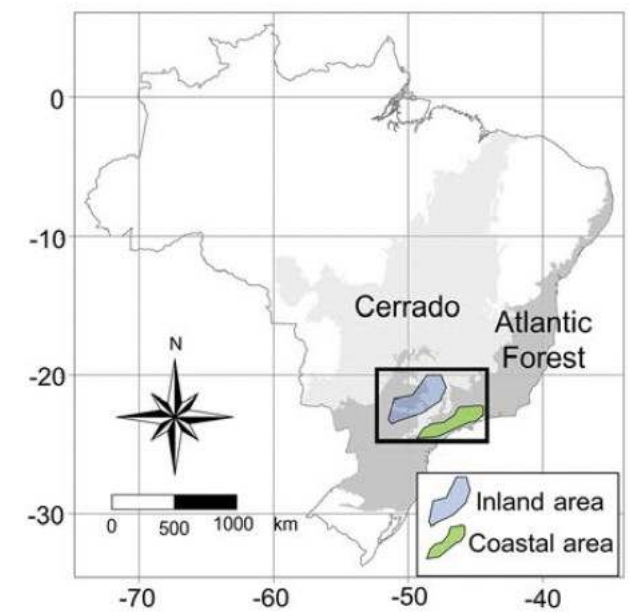
- Uplifts of the Serra do Mar mountain range impacted speciation and dispersal rate of lineages
- Speciation rate of costal areas was double that of inland communities



Differential speciation rates, colonization time and niche conservatism affect community assembly across adjacent biogeographical regions

Ronildo A. Benício^{1,2} | Diogo B. Provete³ | Mariana L. Lyra⁴ | Jani Heino⁵ | Célio F. B. Haddad⁴ | Denise de C. Rossa-Feres⁶ | Fernando R. da Silva⁷

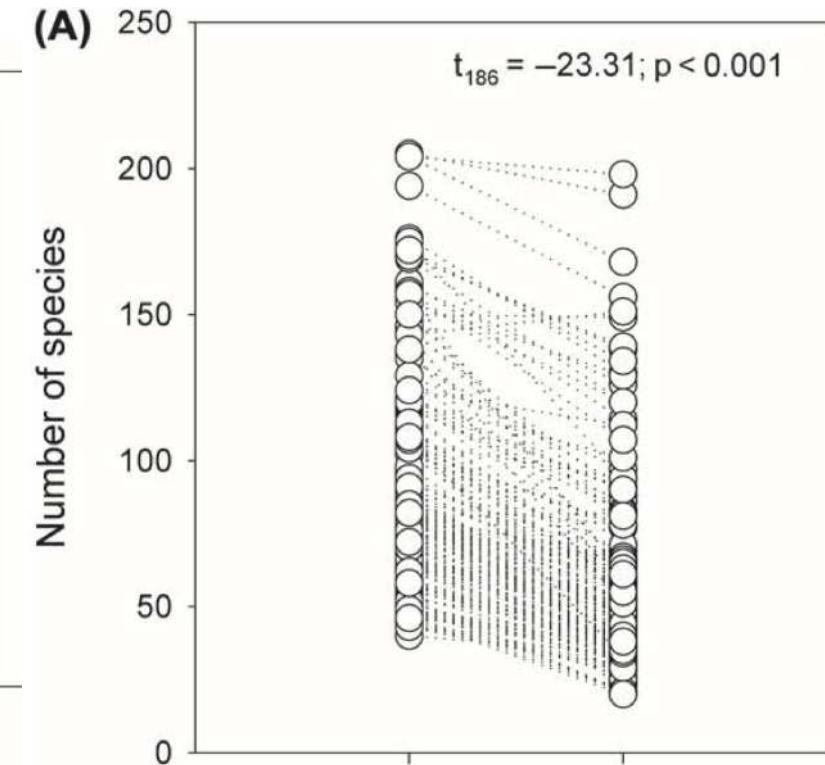
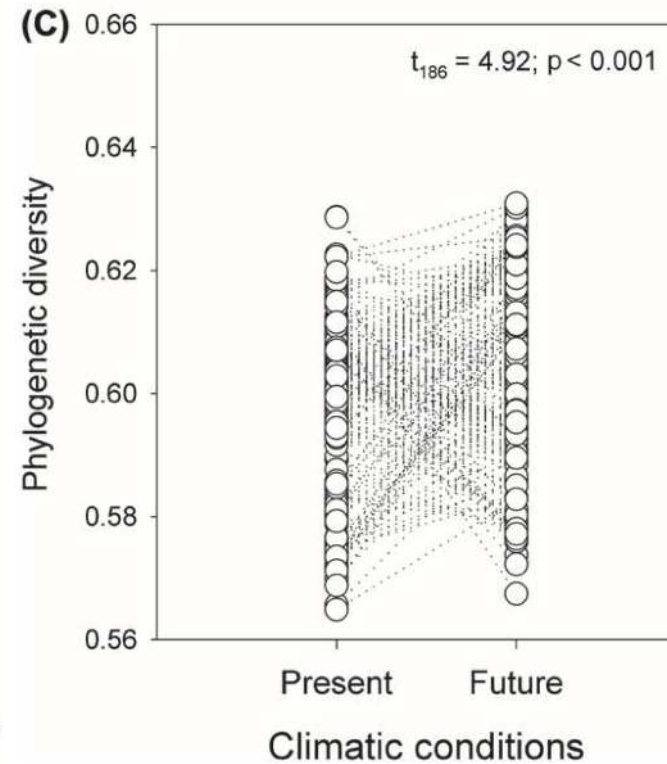
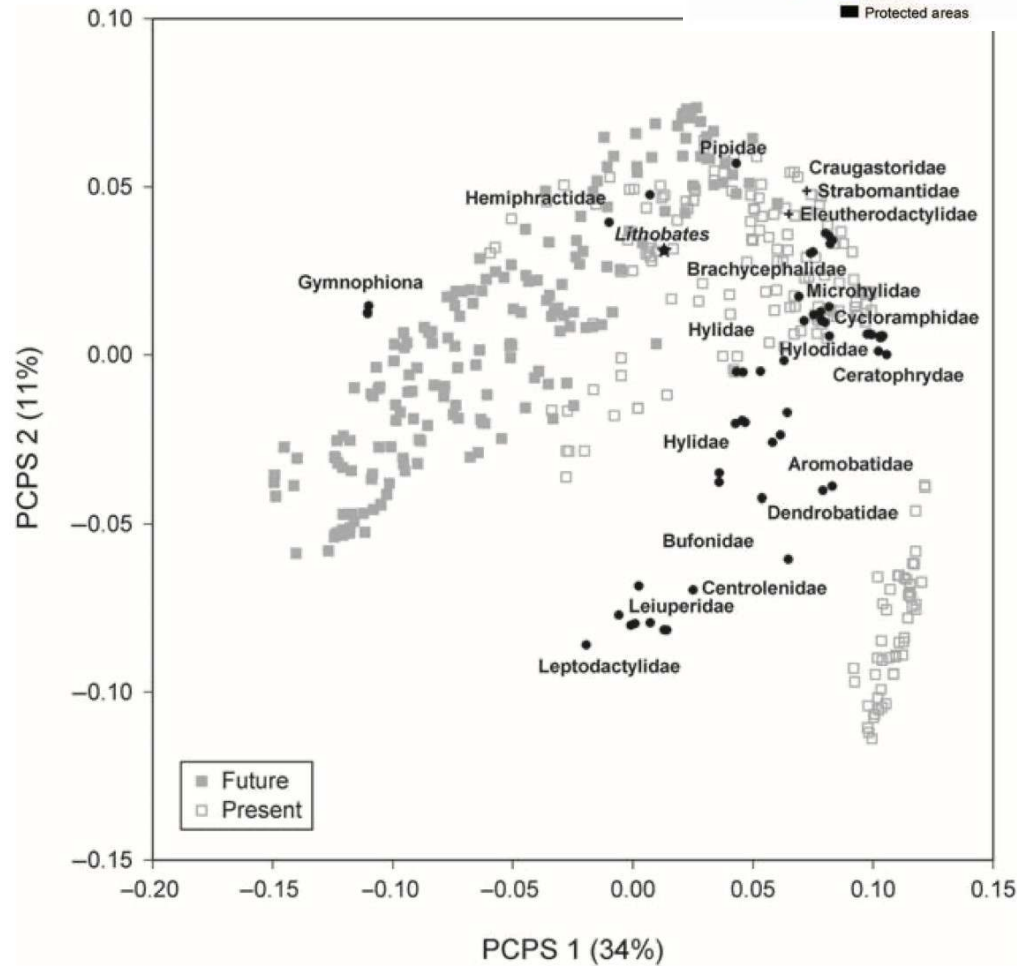
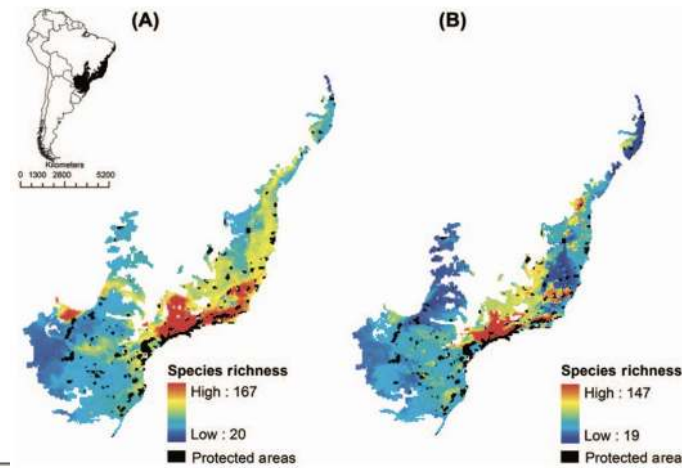
(b) STUDY AREA





Clade-specific consequences of climate change to amphibians in Atlantic Forest protected areas

Rafael D. Loyola, Priscila Lemes, Fernanda T. Brum, Diogo B. Provete and Leandro D. S. Duarte





**Matheus T Moroti
(former PhD student)**



Journal of Biogeography

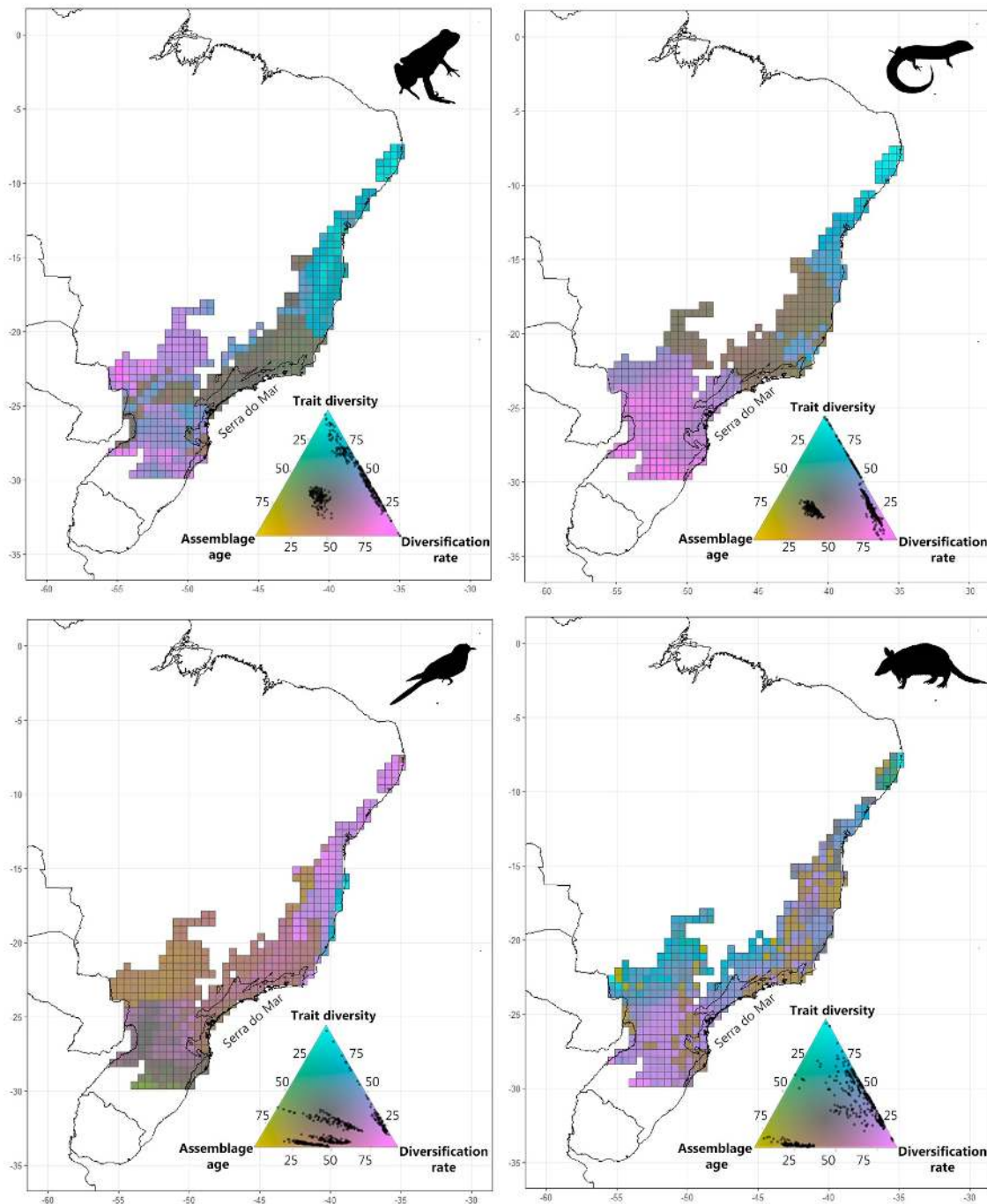
1st round of review

**Climate interacts with diversification rate in determining
species richness and trait diversity of tetrapods in a global
hotspot**

5

6 Matheus de Toledo Moroti^{1,2}, Alexander Skeels^{3,4}, Fernando R. Da Silva⁵, Diogo B.

7 Provete^{2,6*}

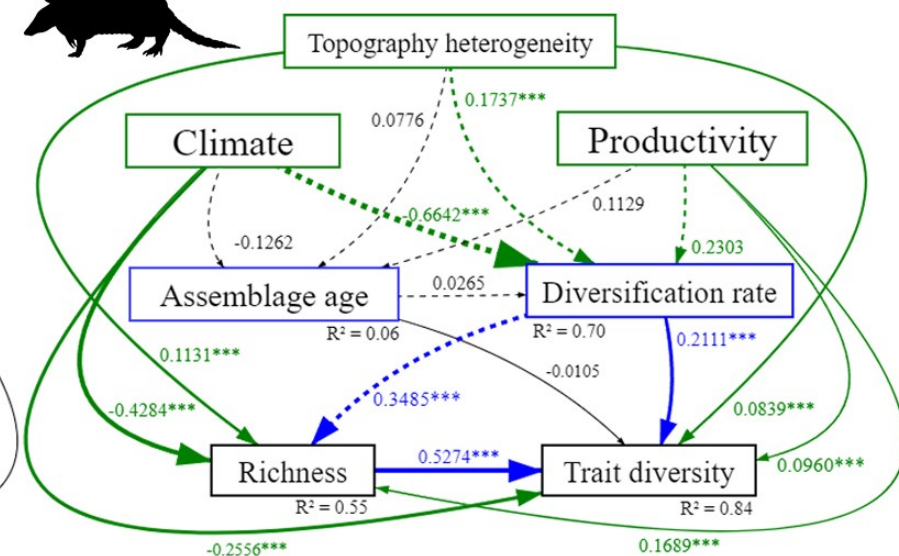
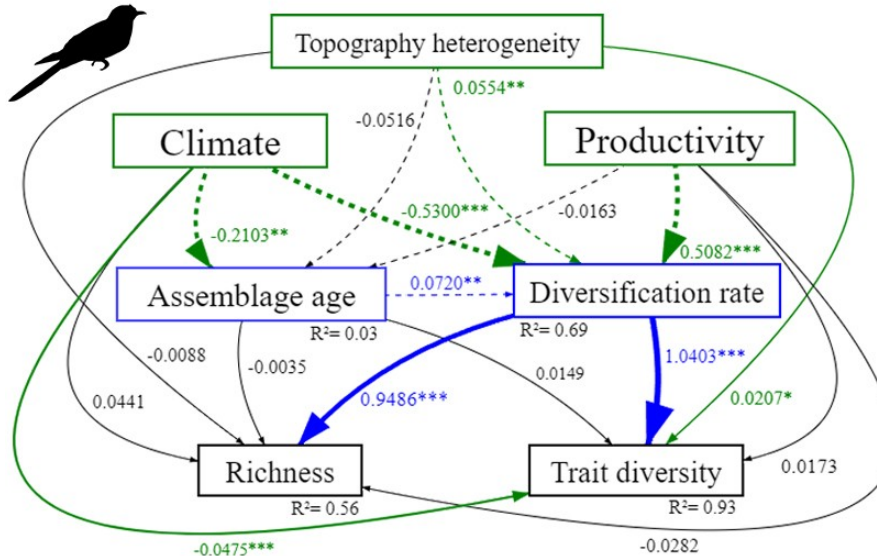
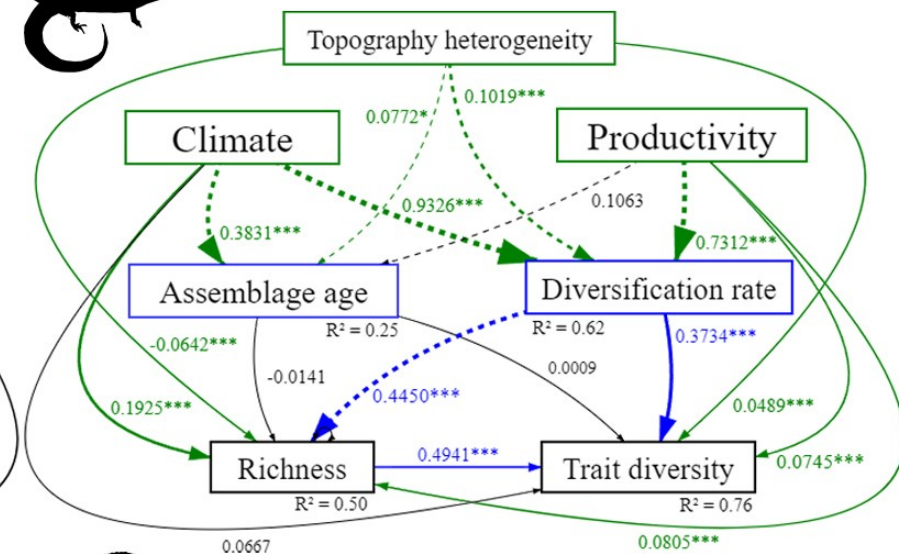
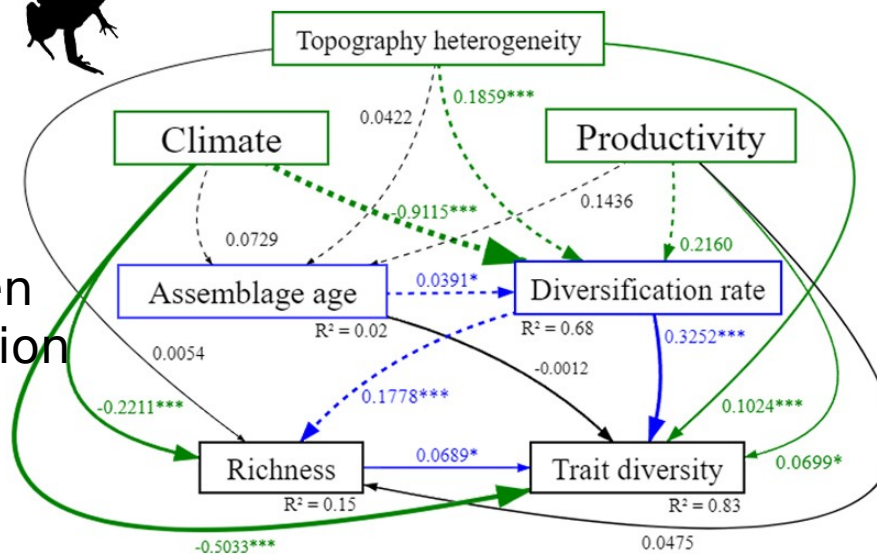


- For frogs and squamates, the oldest communities, with high diversification rate and trait diversity, were in the central portion of the Atlantic Forest
- Oldest communities for Birds were in the central portion, they had highest diversification rates in the transition zones with the Cerrado

Richness	Frogs	Squamates	Birds
Squamates	0.387	-	-
Birds	0.655	0.549	-
Mammals	0.700	0.703	0.776

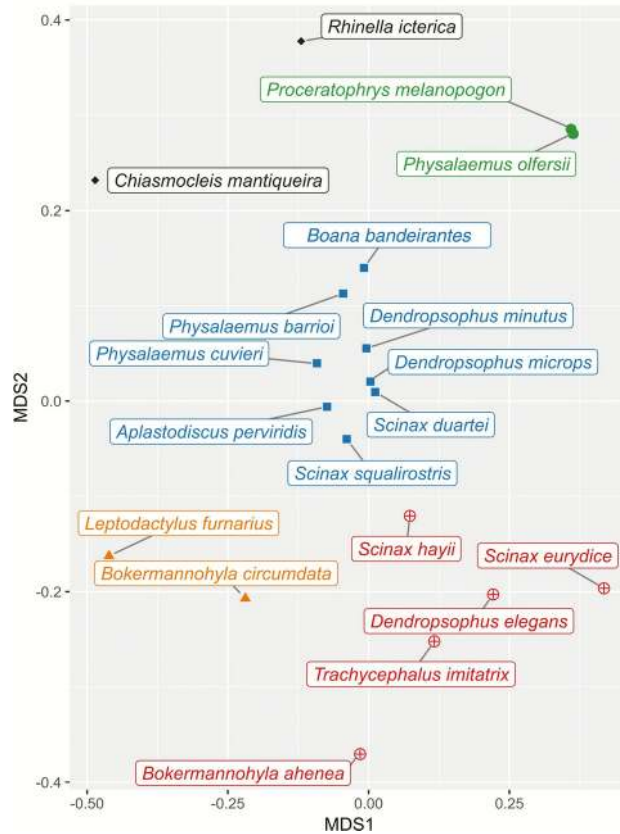
Congruent species richness pattern between most groups, but idiosyncratic ones for trait diversity, assemblage age, and diversification rate

positive relationship between productivity and diversification rate for all groups

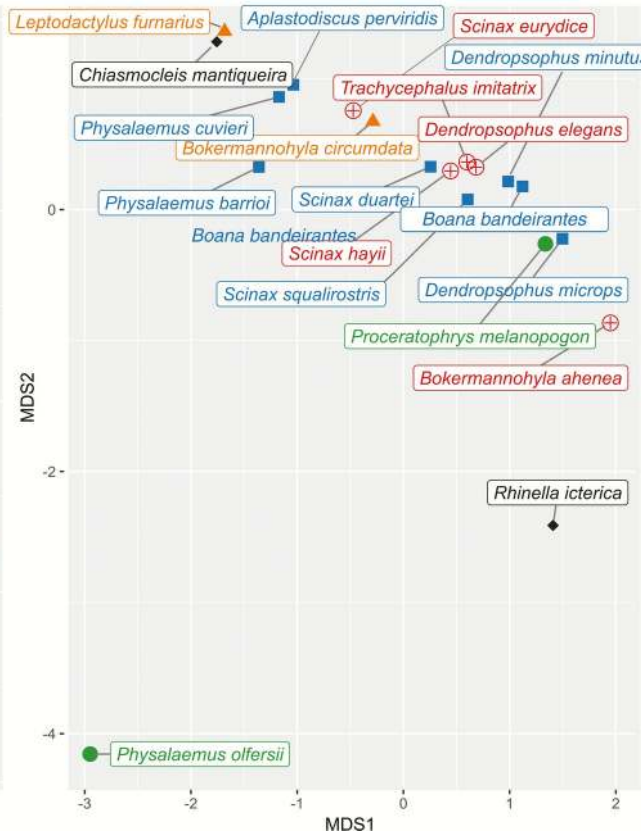


How phenotypic traits influence
(meta)community assembly under an
evolutionary perspective

A)



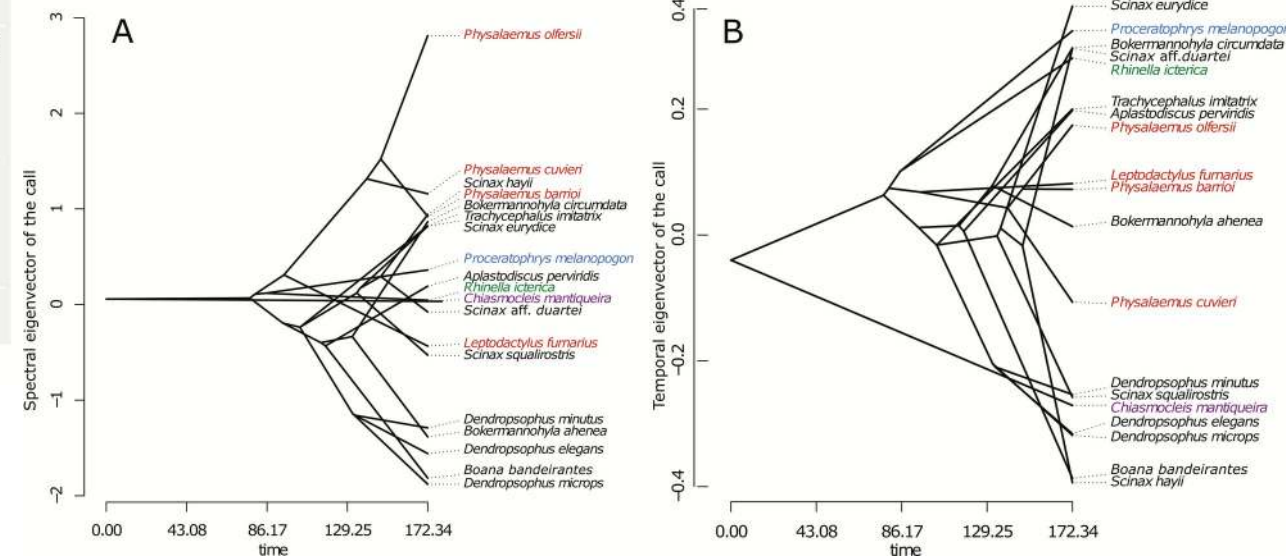
B)



Phylogenetic and adaptive components of the anuran advertisement call correlate with temporal species co-occurrence

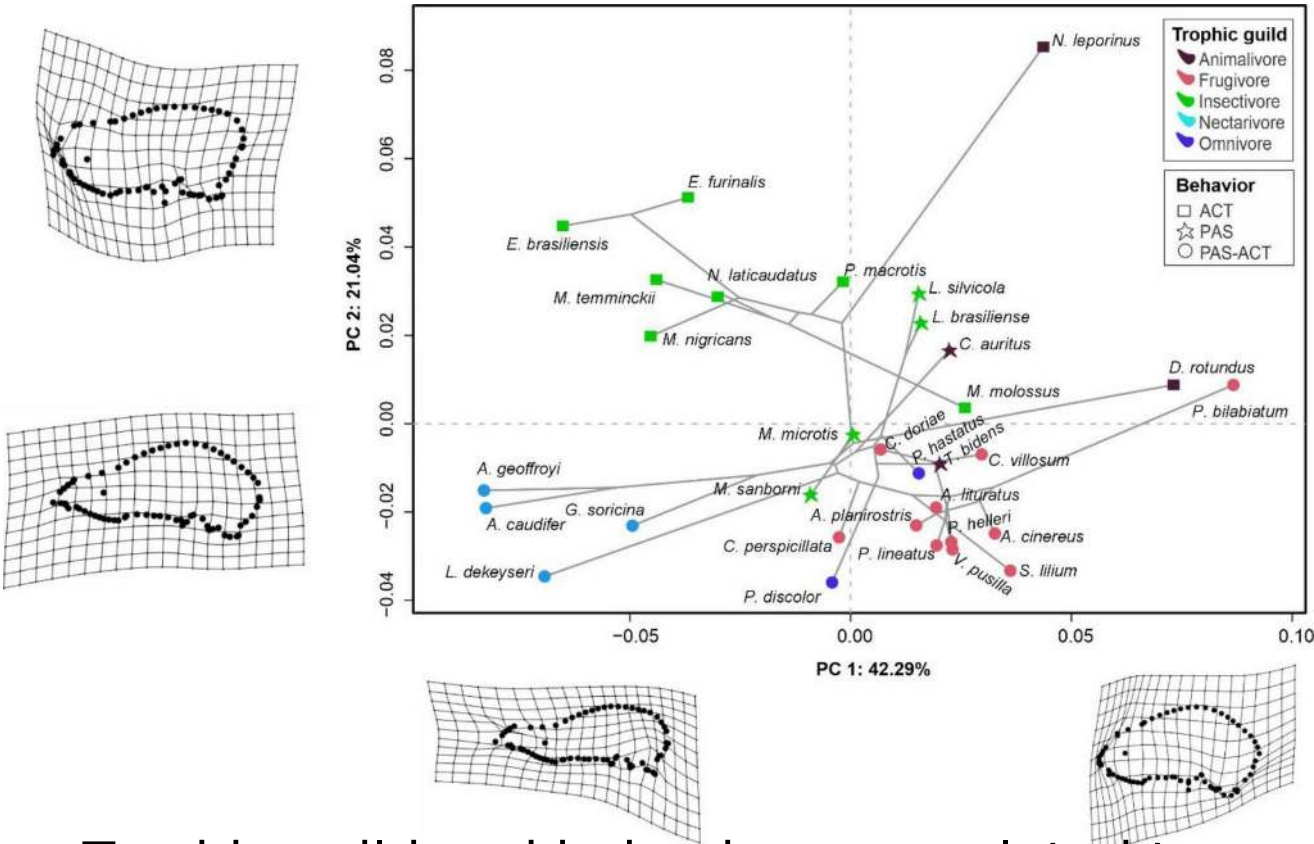
MICHEL VARAJÃO GAREY^{1†*}, DIOGO B. PROVETE^{2†}, THIAGO GONÇALVES-SOUZA³, LILIAN SAYURI OUCHI-MELO⁴, CÉLIO F. B. HADDAD⁵ and DENISE C. ROSSA-FERES⁶

Species occupy distinct regions of the advertisement call space, when considering temporal or spectral parameters



Species with similar temporal co-occurrence patterns have different calling parameters

Spectral parameters (e.g., dominant frequency) are more conserved throughout evolution



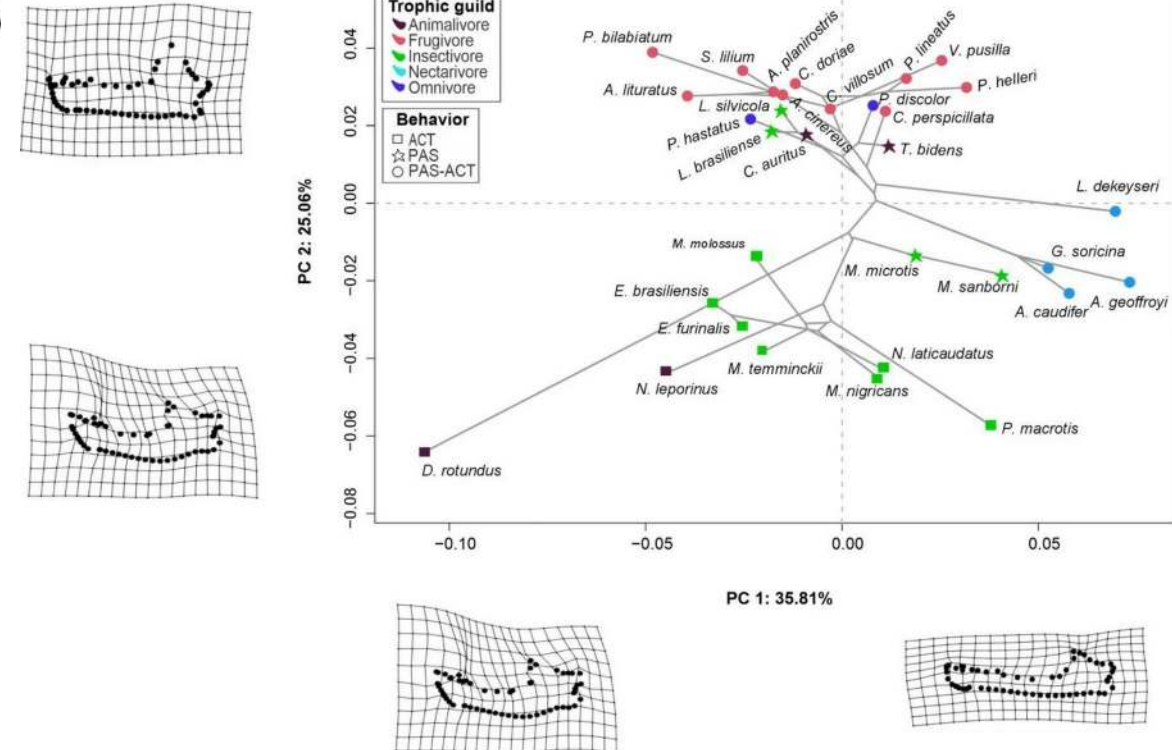
RESEARCH



An evolutionary ecomorphological perspective on the assembly of a neotropical bat metacommunity

Adriana C. Acero-Murcia¹ · Marcos R. Severgnini¹ · Erich Fischer² · Diogo B. Provete^{2,3}

- Trophic guild and behaviour are related to skull shape
- Shape had phylogenetic signal and followed BM model
- Phylogeny, which is correlated with resource use, helps to inform species co-occurrence
- Helps explain why there are so few generalists in species-rich Neotropical bat communities



Take home messages

- Biodiversity in South America, and particularly the Atlantic Forest, seems to have been deeply influenced by mountains, not only the Andes but also others
- The continent harbours important hotspots (e.g., Atlantic Forest, Cerrado), but also coldspots. Both should be considered in conservation prioritization schemes
- At least for frogs, diversification rate seems related to productivity and time-for-speciation effect
- Incorporating an evolutionary perspective helps understanding metacommunity assembly processes